



Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **APT:** Arbitrage Pricing Theory
- ▣ **BAB:** Betting Against Beta
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BH:** Benjamini–Hochberg (procedure)
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAGR:** Compound Annual Growth Rate
- ▣ **CAPM:** Capital Asset Pricing Model
- ▣ **CMA:** Conservative Minus Aggressive (investment factor)
- ▣ **CML:** Capital Market Line
- ▣ **ETF:** Exchange-Traded Fund
- ▣ **FDR:** False Discovery Rate
- ▣ **FF3:** Fama–French three-factor model
- ▣ **FF5:** Fama–French five-factor model
- ▣ **FWER:** Family-Wise Error Rate
- ▣ **GLS:** Generalised Least Squares
- ▣ **GMM:** Generalized Method of Moments
- ▣ **GRS:** Gibbons–Ross–Shanken (test)
- ▣ **HAC:** Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HML:** High Minus Low (value factor)

Acronyms used in this material (2/2)

- ▣ **IPCA**: Instrumented Principal Component Analysis
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **MOM**: MOMentum factor
- ▣ **NYSE**: New York Stock Exchange
- ▣ **OLS**: Ordinary Least Squares
- ▣ **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- ▣ **PC1**: first Principal Component
- ▣ **PC2**: second Principal Component
- ▣ **PC3**: third Principal Component
- ▣ **PCA**: Principal Component Analysis
- ▣ **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- ▣ **RF**: Risk-Free rate
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **RMW**: Robust Minus Weak (profitability factor)
- ▣ **ROE**: Return On Equity
- ▣ **RP-PCA**: Risk-Premium Principal Component Analysis
- ▣ **SDF**: Stochastic Discount Factor
- ▣ **SMB**: Small Minus Big (size factor)
- ▣ **SML**: Security Market Line
- ▣ **UMD**: Up Minus Down (momentum factor)
- ▣ **US**: United States

Chapter Route and Outcomes

Four questions

- ▣ What explains expected returns? The CAPM and multifactor models
- ▣ How are exposures estimated? Betas, their uncertainty and shrinkage
- ▣ How are pricing restrictions tested? GRS, Fama–MacBeth, model comparison
- ▣ Do the findings survive? Multiple testing, latent factors, replication

After this chapter you can

- ▣ estimate a beta with HAC errors and shrink it
- ▣ test the CAPM jointly and estimate risk premia with robust errors
- ▣ judge a new factor against multiple-testing hurdles and replication evidence

Part I: Sections 1–5; Part II: Sections 6–13; technical detail in the [Appendix](#)

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \dots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $R_{i,t}$: return of asset i in period t ; $R_{f,t}$: risk-free rate; K : number of factors; α_i : average return not explained by the factors, 0 if they price the asset
- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k ; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock
- Two uses in this chapter
 - ▶ Explain an asset: the technology ETF XLK moves 1.28 times the US market (Section 3)
 - ▶ Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Also: cost of capital, risk management (N assets need $N(N+1)/2$ covariances, K factors far fewer), portfolio construction (Campbell, Lo & MacKinlay (1997), ch. 5–6)
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

Data and Conventions

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size $\times$ B/M portfolios	US stocks, two sorts	1963–2026
US sector ETFs	S&P 500 sectors: 9 from 1999, 11 from 2018	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 stocks and their index	2015–2026

- ▣ Factors and portfolios: [Kenneth French Data Library](#), in percent (divide by 100); ETFs and stocks: daily adjusted closing prices (BET: closing values), to 18 Sep 2026
- ▣ **Excess return** $R_{i,t}^e = R_{i,t} - R_{f,t}$; $R_{f,t}$: one-month Treasury bill rate (RF in the French data)
- ▣ **B/M**: book value of equity over its market value; **BET**: the benchmark index of the BVB
- ▣ Simple returns for regressions on the French factors; log returns for rolling, BVB and PCA analyses
- ▣ BVB: daily $R_{f,t}$ treated as constant; it cancels in covariances, so betas are unchanged

Mean–Variance and the Tangency Portfolio

- **Markowitz (1952)**: with weights w , expected excess returns μ and covariance Σ , the frontier solves

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, w^\top \mathbf{1} = 1$$

- ▶ in words: the lowest variance for a target mean μ^* ; $\mathbf{1}$: a vector of ones, so the weights sum to 1; $^\top$: transpose

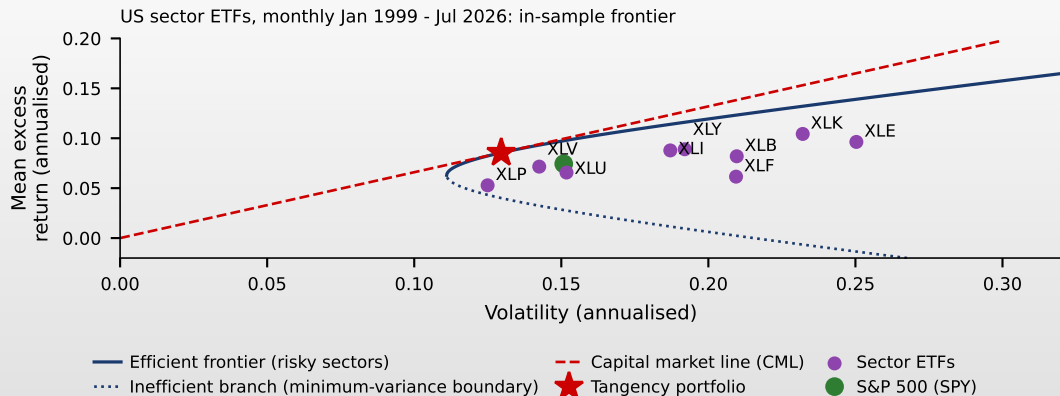
- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f]/\sigma_p$; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$w_{\text{tan}} = \Sigma^{-1} \mu / \mathbf{1}^\top \Sigma^{-1} \mu$$

- ▶ R_p, σ_p : return and volatility of portfolio p ; the denominator rescales $\Sigma^{-1} \mu$ so that the weights sum to 1

- **CML**: mixes of R_f and the tangency portfolio, $\mu_p = SR_{\text{tan}} \sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A2; estimation error and shrinkage: [Chapter 4](#)

The Efficient Frontier of US Sectors (1/2)



The Efficient Frontier of US Sectors (2/2)

- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the S&P 500 ETF, the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49: here the market is **not** on the frontier, as the CAPM would require
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- The frontier is **in-sample**; estimated means are very noisy ([Chapter 4](#))

 MFM_ch3_frontier

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in
Economics 1990

Photo: Larry D. Moore (2007); CC BY
4.0; Wikimedia Commons

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- If every investor holds a mean–variance efficient portfolio, all hold the same tangency portfolio, which must then be the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- R_i, R_m : returns of asset i and of the market portfolio; $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i : sensitivity of asset i to the market, $\beta_i > 1$ amplifies market moves
- [Black \(1972\)](#): a version without a risk-free asset, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

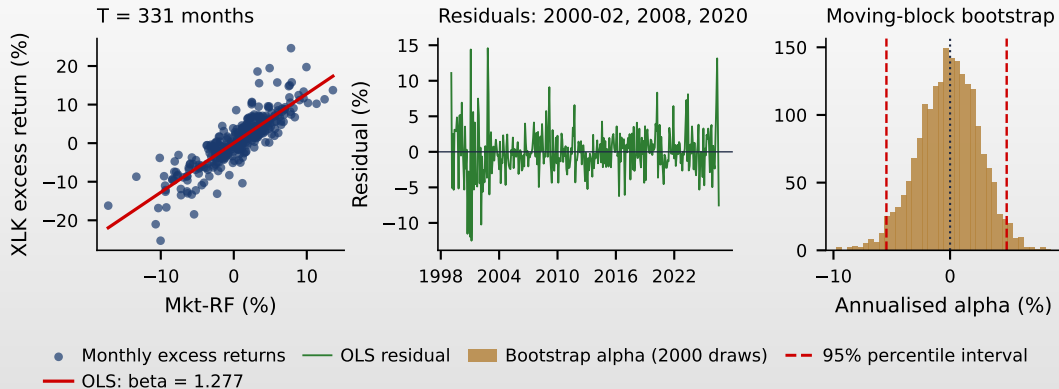
What the CAPM Implies

- Main assumptions
 - ▶ Mean–variance investors with identical beliefs, one period
 - ▶ Unlimited borrowing and lending at R_f ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns; idiosyncratic risk earns nothing
- **SML** (Security Market Line): the testable statement of the CAPM (Section 4)
 - ▶ every asset lies on the line (beta, mean), with slope equal to the market premium
 - ▶ the CML contains only efficient portfolios
- Cost of capital from the CAPM needs an estimated beta and an estimated premium: the error of the premium dominates
- Which portfolio is “the market”? The answer decides what a test of the CAPM can establish (Roll’s critique, Section 4)

The Market Model, HAC Errors and Estimation Choices

- ▣ Market model $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\text{Cov}}(R_i^e, R_m^e) / \widehat{\text{Var}}(R_m^e)$
- ▣ Variance split $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
 - ▶ σ_i^2, σ_m^2 : variances of the asset and of the market; $\sigma_{\varepsilon,i}^2$: residual variance; $R^2 = \beta_i^2 \sigma_m^2 / \sigma_i^2$
- ▣ Volatility clustering ([Chapter 1](#)) makes residuals heteroskedastic: use **HAC** errors ([Newey & West \(1987\)](#))
 - ▶ Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags; T : number of periods, $\lfloor \cdot \rfloor$: integer part
 - ▶ $T = 331$: $L = 5$; $T = 757$: $L = 6$; the point estimates are those of OLS
- ▣ Choices to report with every beta
 - ▶ Frequency and window: 3–5 years of monthly or 1–2 years of daily data; daily data suffer from asynchronous trading
 - ▶ Market proxy: SPY (S&P 500) for US assets, BET for Romanian stocks
 - ▶ Data cleaning: an unadjusted corporate event creates a fake return (DIGI, May 2017: a daily log return of -6.6)

Worked Example: Beta of the Technology ETF (1/2)



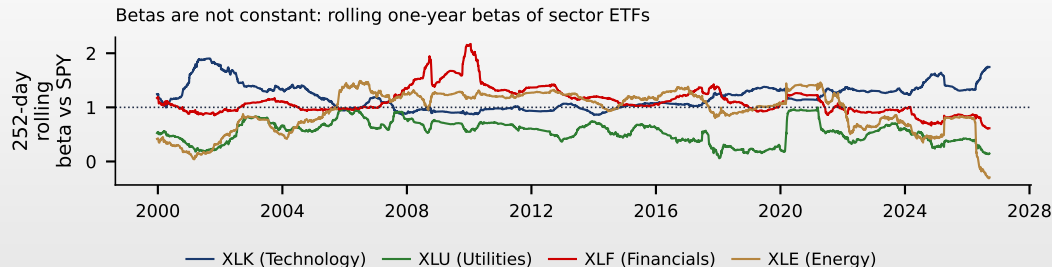
Worked Example: Beta of the Technology ETF (2/2)

	Estimate	Classical	White	Newey–West
α (ann.)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly 1999–2026, $T = 331$, $R^2 = 0.742$
- Robust s.e. of β : twice the classical one; heteroskedasticity explains part (White), serial correlation of $f_t \varepsilon_t$ the rest
- $t(\alpha) = 0.33/2.45 = 0.13$; bootstrap interval $[-5.5\%, 4.9\%]$

 MFM_ch3_seminar

Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end
- Energy (XLE): from -0.31 to 1.49; negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes; conditional models: [Chapters 5–6](#)

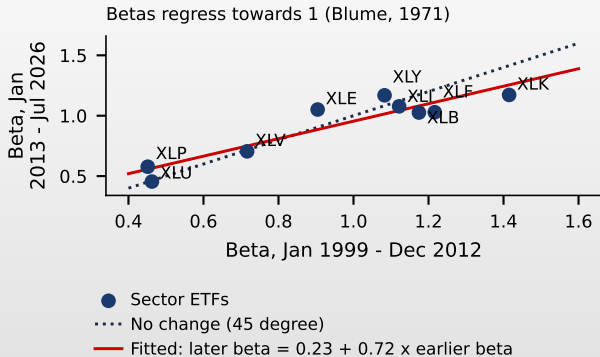
Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is $\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior mean $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

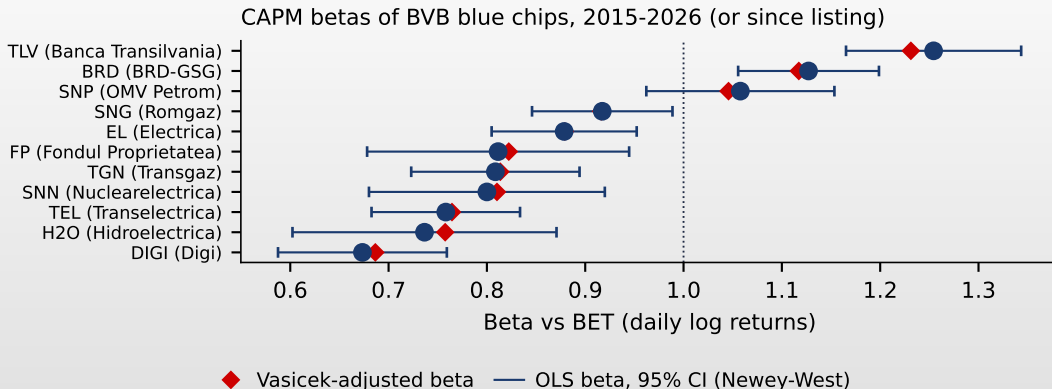
- ▣ w_i : the weight of the estimate, larger when $\hat{\beta}_i$ is precise (small se); $\text{se}(\hat{\beta}_i)$: its standard error
- ▣ σ_{β}^2 is the variance of the **true** betas (Var_{cs} : variance across assets; $\overline{\text{se}^2}$: mean squared standard error):
 $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$ (posterior derivation: Seminar 3, problem A3)
- ▣ XLK, monthly 1999–2012, 9 sectors: $\hat{\beta} = 1.415$ (se 0.139), $\bar{\beta} = 0.949$, $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$, $w = 0.849$
 ► $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$; Blume 1.278; beta estimated on 2013–2026: 1.172

Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67 \hat{\beta}$
- **RMSE** of the forecast of later betas, $\sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$: raw 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%; with 9 sectors the lesson is qualitative

CAPM Betas on the Bucharest Stock Exchange (1/2)



CAPM Betas on the Bucharest Stock Exchange (2/2)

- Daily log returns on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023); days with $|r_t| > 0.5$ removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD 1.13; OMV Petrom 1.06; Transelectrica 0.76, Digi 0.67
- R^2 : 0.59 for TLV, only 0.21 for DIGI; large index weights raise R^2 mechanically
- Vasicek changes little: the daily estimates are precise (se ≈ 0.04 – 0.07); stale prices bias daily betas of less liquid stocks (Seminar 3, problem B4)

 MFM_ch3_bvb_betas

From One Alpha to the GRS Test (1/2)

- ▣ **Jensen's alpha** (Jensen (1968))
 - ▶ the market-model intercept α_i ; the CAPM implies $\alpha_i = 0$ for every asset
- ▣ One t -test per asset (XLK: $t = 0.13$)
 - ▶ with many assets, false rejections accumulate (Section 8)
- ▣ **GRS** (Gibbons, Ross & Shanken (1989)): a joint test of $H_0 : \alpha_1 = \dots = \alpha_N = 0$

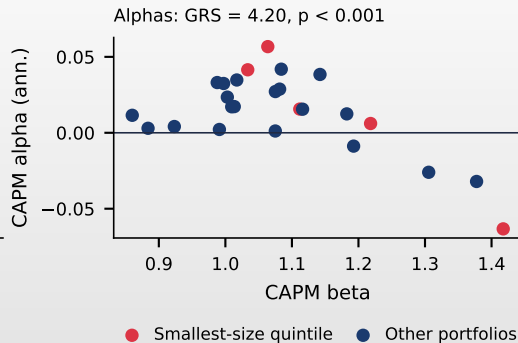
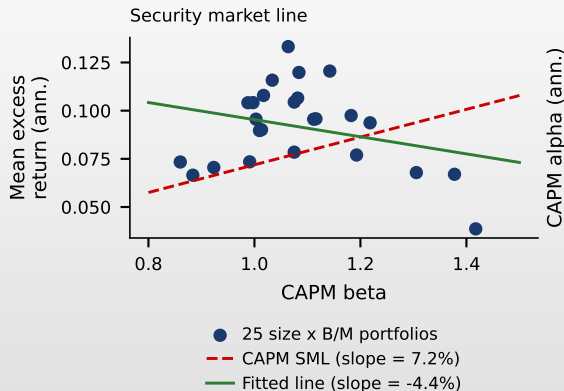
$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣ Notation
 - ▶ N : number of test assets; T : number of months; $\hat{\alpha}$: the N estimated alphas
 - ▶ $\hat{\Sigma}$: maximum-likelihood residual covariance (divided by T)
 - ▶ $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market

From One Alpha to the GRS Test (2/2)

- ▣ $\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}$: the alphas squared and weighted by the inverse residual covariance
- ▣ Decision: reject H_0 when W exceeds the 95% quantile of $F(N, T - N - 1)$; F : Fisher distribution
- ▣ The distribution is exact, conditional on the factors
 - ▶ errors i.i.d. multivariate Normal, independent of the factors
 - ▶ zero mean and constant nonsingular covariance
- ▣ Large W : the market is far from the mean–variance frontier of the test assets

The Security Market Line in the Data (1/2)



The Security Market Line in the Data (2/2)

- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026, $T = 757$
- The CAPM line has slope 7.2% a year; the fitted line has slope -4.4% : higher beta, **lower** average return
- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Small-value alpha $+5.7\%$, small-growth -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- $GRS = 4.20$, p -value 8×10^{-11} : the CAPM is rejected

 MFM_ch3_sml

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\Sigma}$ and the moments estimated by maximum likelihood

$$\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2(f, R) - \widehat{SR}^2(f)$$

- $\widehat{SR}^2(f, R)$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets; $\widehat{SR}^2(f)$: that of the factor alone

- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T - N - 1}{N} \cdot \frac{\widehat{SR}^2(f, R) - \widehat{SR}^2(f)}{1 + \widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1 + 0.134^2) = 4.20$; the gain expected under H_0 is far smaller ([Appendix](#))

When the GRS Assumptions Fail

- **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
 - ▶ Residual bootstrap under H_0 : resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B7)
 - ▶ the bootstrap is robust to non-Normality only if the errors are i.i.d. and independent of the factors
 - ▶ GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product)
 - ▶ $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_N^2$, $\hat{V}_\alpha$: HAC estimate of $\text{Var}(\hat{\alpha})$; robust to heteroskedasticity and autocorrelation
- **Rank condition**: $\hat{\Sigma}$ has rank at most $T - 2$
 - ▶ GRS is not computable when $N \geq T - 1$, and its power falls well before that point
- Tests built for many assets: [▶ Appendix](#)

A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia
(2009 upload); public domain; Wikimedia
Commons

- **Black (1972)**: if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage; high-beta assets become expensive and earn less
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above R_f
- **Frazzini & Pedersen (2014)** turn this into a strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L : low-beta portfolio leveraged to beta 1; H : high-beta portfolio de-leveraged to beta 1; BAB has beta zero by construction and a positive average return in their data
- An unscaled low-minus-high-beta spread (Seminar 3, problem C2) is not beta-neutral: its mean mixes the premium with a negative market exposure

What a CAPM Test Establishes

- **Roll (1977)**: the CAPM is a statement about the **true** market portfolio of all wealth; stock indices omit bonds, real estate, private firms and human capital
- Any efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (**Black (1972)**) on the risky-asset frontier (Seminar 3, problem A2)
 - ▶ A CAPM test is a joint test of the model and of the efficiency of the proxy
- **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ($T = 331$): $GRS = 0.53$, $p = 0.85$, not rejected
 - ▶ On the same 331 months the 25 size $\times$ B/M portfolios still reject: $GRS = 2.91$, $p = 9 \times 10^{-6}$
 - ▶ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- Failing to reject is weak evidence; evaluate models by how well they price assets with dispersion in expected returns

Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a K -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣ β_i : the K loadings of asset i ; f_t : the factor surprises, with mean 0; λ : the **risk premia**, the extra expected return per unit of each exposure
- ▣ The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \frac{1}{3}(S_L + S_N + S_H) - \frac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \frac{1}{2}(S_H + B_H) - \frac{1}{2}(S_L + B_L)$$

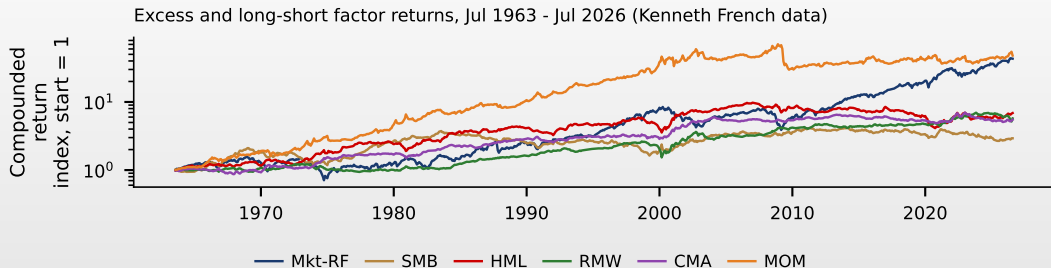
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread,
 $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French 2×3 sort each June, accounting data of the previous year, value-weighted portfolios
 - ▶ **SMB** (Small Minus Big): small minus big stocks
 - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart q-factor	FF3 + MOM market, size, investment, ROE	Carhart (1997) Hou, Xue & Zhang (2015)	fund evaluation firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3: $R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$, with $\text{MKT} = \text{Mkt-RF}$ and loadings b_i, s_i, h_i on the three factors; a good model leaves time-series alphas close to zero (GRS not rejected)
- Factors beyond FF3 (ROE: Return On Equity)
 - ▶ RMW (Robust Minus Weak): high minus low operating profitability
 - ▶ CMA (Conservative Minus Aggressive): low minus high asset growth
 - ▶ MOM (UMD, Up Minus Down): high minus low return from month $t - 12$ to $t - 2$; past winners keep winning over 3–12 months (Jegadeesh & Titman (1993))
- FF3 uses its own SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from three sorts, so FF3 is not nested in FF5

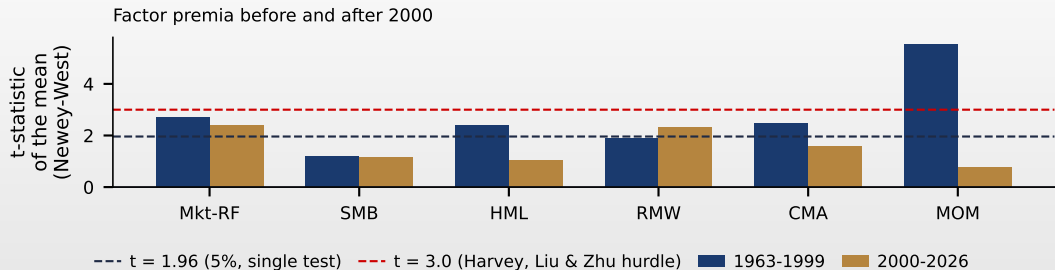
Six Factors over Sixty-Three Years



- Compounded monthly factor returns from 1, log scale: excess or long-short returns, so no interest on capital or collateral; not the wealth of a funded investment
- July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- Value was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

 MFM_ch3_factor_cum

Do the Premia Survive?



- t -statistic of the mean monthly factor return (Newey-West), 1963–1999 vs 2000–2026
- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31; the market premium holds in both halves ($t = 2.70$ and 2.39)
- A calendar split is not a publication event: dating the decay needs the design of Section 12

The Two Passes



Eugene Fama, Morgan Stanley–AFA
Award, 2008

Photo: TonyTheTiger (2008); CC

BY-SA 3.0; Wikimedia Commons

- Fama & MacBeth (1973) estimate the risk premia λ of $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$ (Fama: Nobel Prize 2013)

- **Pass 1:** time series, one regression per asset

- ▶ returns $T \times N$, factors $T \times K \Rightarrow$ betas $\hat{B}$, $N \times K$

$$R_{i,t}^e = a_i + \beta_i^\top f_t + \varepsilon_{i,t}$$

- **Pass 2:** cross section, one regression per month

- ▶ the N returns on $X = [1, \hat{B}]$

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \lambda_t + u_{i,t}, \quad \hat{\lambda} = \frac{1}{T} \sum_t \hat{\lambda}_t$$

- λ_0 : pricing error common to all assets, zero if the model is right; a_i : pass-1 intercept; $u_{i,t}$: pass-2 residual; $\hat{\lambda}$: average of the monthly slopes
- Allows factors that are **not** traded portfolios, and firm characteristics as regressors

Standard Errors for Risk Premia (1/2)

- **Fama–MacBeth:** the monthly $\hat{\lambda}_{k,t}$ form a sample of size T

$$se_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$$

- ▶ $s(\cdot)$: the sample standard deviation over the T months
- ▶ robust to cross-sectional correlation of the residuals
- ▶ if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- **Misspecification:** the Fama–MacBeth, Newey–West and Shanken (next slide) errors assume a correct model
 - ▶ pricing errors $e = \mu - X\gamma = 0$; μ : mean excess returns, $\gamma = (\lambda_0, \lambda^\top)^\top$
 - ▶ when GRS rejects, the noise in $\hat{B}$ multiplies e and adds variance (Kan, Robotti & Shanken (2013); GMM form in the [Appendix](#))

Standard Errors for Risk Premia (2/2)

- ▣ **Errors in variables:** pass 2 uses $\hat{\beta}$, not β , so se_{FM} is too small
- ▣ **Shanken (1992)** scales the non-factor part of the variance by $(1 + c)$

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \left[\text{Var}_{FM}(\hat{\lambda}) - \frac{1}{T} \Sigma_f \right] + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

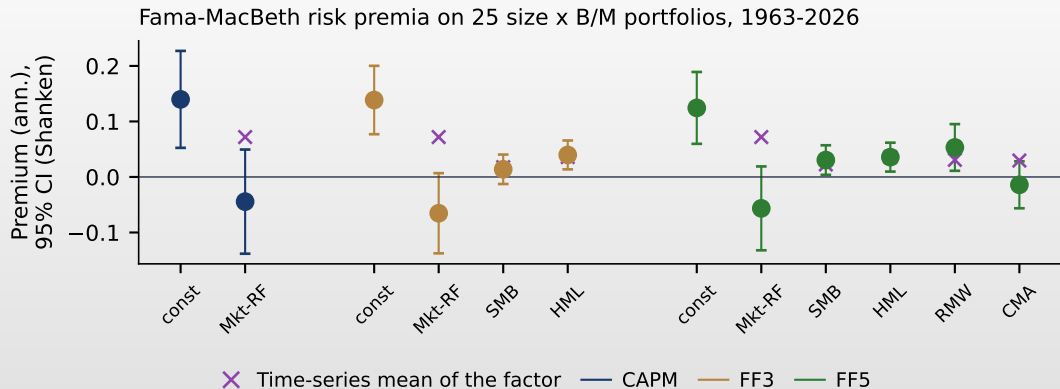
- ▶ Var_{FM} : the Fama–MacBeth variance, $s^2(\hat{\lambda}_t)/T$; Σ_f : factor covariance
- ▶ c : squared Sharpe ratio of the premia per period, 0.03 for FF3 here, larger for some non-traded factors
- ▣ Small c (traded factors, monthly data): the correction is small; large c : Fama–MacBeth t -statistics are overstated

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	s.e.	t	Significant at 5%?
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026; Shanken barely moves the error ($c = 0.03$ per month)
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- FF5: the misspecification-robust s.e. of RMW rises from 2.1% to 2.6%, and its t falls from 2.5 to 2.1
- At the $t > 3$ hurdle of Section 8, HML is borderline under every choice (2.45–3.00)

Estimated Risk Premia (1/2)



Estimated Risk Premia (2/2)

- 25 size $\times$ B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means
- CAPM: intercept 14.0% (se 4.4%), market premium –4.4% (se 4.8%)
- FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); FF5: RMW 5.3% (2.1%), CMA –1.4% (2.2%)
- For a traded factor, λ should equal its time-series mean: HML 4.0% vs 3.6%, close; market (FF3) –6.5% vs 7.2%, far
- Market betas span only 0.86–1.42 (cross-sectional s.d. 0.14): λ_m is **weakly identified**, and a large intercept absorbs the level ([Lewellen, Nagel & Shanken \(2010\)](#))

 MFM_ch3_fama_macbeth

Useless and Weak Factors

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$, of the order of $1/\sqrt{T}$, is pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t -test over-rejects when the model is misspecified ([Kan & Zhang \(1999\)](#))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
 - ▶ $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
 - ▶ First-pass Wald test of $\beta_{i,g} = 0$ for all i : rejected in 6.7%; run it before pass 2
- **Weak factors**: little dispersion in betas, like the market here
 - ▶ identification-robust tests ([Kleibergen \(2009\)](#))
 - ▶ with many assets, consistency of pass 2 depends on factor strength ([Anatolyev & Mikusheva \(2022\)](#))

A Skeptical Appraisal: Adding 30 Industry Portfolios

Model	OLS R^2		GLS R^2	
	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

Monthly 1963–2026, free intercept and premia

- The 25 portfolios have a strong three-factor structure: any factors correlated with SMB and HML fit their means, so a high OLS R^2 is a low hurdle ([Lewellen, Nagel & Shanken \(2010\)](#))
- Three checks
 - ▶ Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
 - ▶ Report the **GLS** (Generalised Least Squares) R^2 , weighted by $\hat{V}^{-1}$ like GRS: never above 0.26
 - ▶ Impose $\lambda = \mathbb{E}[f]$ for traded factors, no intercept: FF3 on the 55 assets gives $R^2 = -0.27$
- Stocks instead of portfolios: more dispersion in betas, larger errors in variables ([Ang, Liu & Schwarz \(2020\)](#))

Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in
Economics 2013

Photo: Bengt Nyman (2013); CC BY
2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor) m_t : prices all excess returns, $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF, $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$; $\mathbf{b}$: the SDF loadings, $\mathbb{E}\mathbf{f}$: the factor means
- Substitute: $\mathbb{E}[R_i^e] = \text{Cov}(R_i^e, \mathbf{f}_t)^\top \mathbf{b} = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ with $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathbf{b}$: the beta pricing model of Section 6
- $\mathbf{b} \neq 0$ asks whether a factor helps price the assets; $\boldsymbol{\lambda} \neq 0$ whether it is rewarded
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#); survey of the discount-rate view: [Cochrane \(2011\)](#)

Comparing Models: Maximum Sharpe Ratios

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(f) = \mu_f^\top \Sigma_f^{-1} \mu_f$ of its factors (μ_f , Σ_f : means and covariance of the factors), the squared Sharpe ratio of their tangency portfolio; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha of the added factor on the others (spanning regression)
 - ▶ HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994))
- Another model-independent metric, the Hansen–Jagannathan distance: ▶ Appendix

The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs $|t| > 3.0$

Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- **FWER** (Family-Wise Error Rate): probability of at least one false rejection ([Chapter 2](#))
- Notation of the table: M : number of tests; $p_{(1)} \leq \dots \leq p_{(M)}$: sorted p -values; $H_{(j)}$: the hypothesis with the j -th smallest; **BH**: the Benjamini–Hochberg procedure

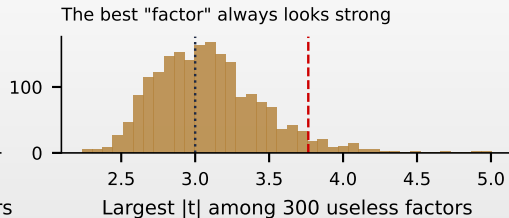
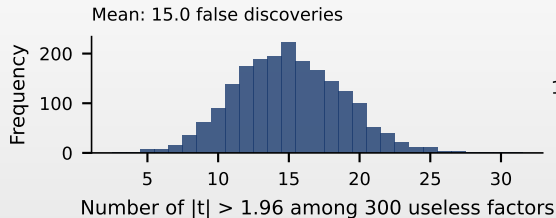
Procedure	Controls	Reject $H_{(j)}$ if	Valid under
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M - k + 1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq \text{largest } k \text{ with } p_{(k)} \leq k\alpha/M$	independence, positive dep.
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=1}^M 1/k$	any dependence

- Positive dependence: the test statistics of the true nulls move up together (PRDS condition, [Benjamini & Yekutieli \(2001\)](#))
- FDR control suits a search over many candidate factors

Worked Example: Ten Candidate Factors

- ▣ Sorted p -values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; $\alpha = 5\%$
- ▣ Naive ($p \leq 0.05$): **7** discoveries
- ▣ Bonferroni ($p \leq 0.005$): **2**
- ▣ Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- ▣ BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**
- ▣ Seminar 3, problem A1 repeats the procedure on a different vector, with a threshold chart

Simulation: 300 Useless Factors



-- Bonferroni threshold 3.76 $t = 3.0$

- 2000 replications; each draws 300 independent t -statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications; median largest $|t|$: 3.06
- Bonferroni threshold for 300 tests: $|t| > 3.76$; hurdles rise with the number of tests actually tried, including unpublished ones

 MFM_ch3_false_discoveries

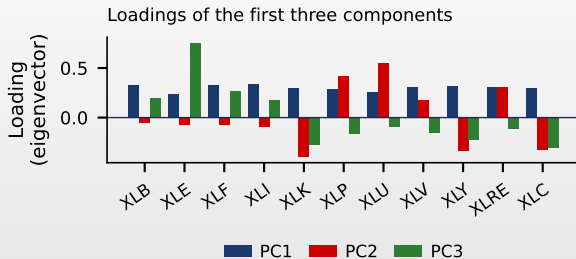
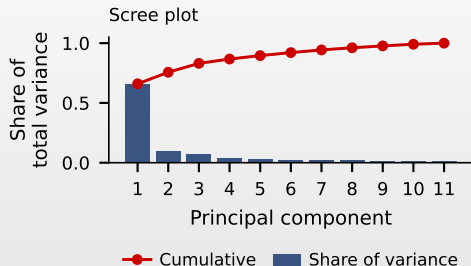
Principal Components in Brief

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis) of the correlation matrix C of standardised returns

$$C = \Gamma \Lambda \Gamma^\top, \quad y_t = \Gamma^\top z_t$$

- ▶ z_t : standardised returns of day t ; y_t : the components
- ▶ Γ : eigenvectors in columns; Λ : diagonal matrix of eigenvalues
- ▣ Eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$; eigenvectors are the **loadings**; component k explains λ_k/N of the variance
- ▣ Two assets with correlation ρ : $\lambda = 1 \pm \rho$, eigenvectors $(1, 1)/\sqrt{2}$ (the “market”) and $(1, -1)/\sqrt{2}$ (the spread); in a crisis ρ rises and PC1 dominates
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components; N equicorrelated assets: Seminar 3, problem A8

PCA of the US Sector ETFs



- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- PC1 explains 66.0%; all loadings positive (0.24–0.34); correlation with SPY 0.954: strongly market-related

- PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology -0.40, consumer discretionary -0.33: **defensive versus growth**, with no name given in advance
- PC3 (7.4%) is dominated by energy; three components explain 83.1%

How Many Factors?

- Approximate factor model $x_{it} = \lambda_i^\top F_t + e_{it}$, $N, T \rightarrow \infty$
 - ▶ x_{it} : standardised return; F_t : factors; λ_i : loadings; e_{it} : idiosyncratic part
 - ▶ $V(k)$: mean squared residual after the first k components; μ_k : the k -th largest eigenvalue
- Bai & Ng (2002): $\hat{k} = \arg \min_k [\ln V(k) + k \frac{N+T}{NT} \ln \min(N, T)]$ (criterion IC_{p2}); Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg \max_k \mu_k / \mu_{k+1}$, no penalty to choose
- Kaiser ($\mu_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size $\times$ B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k : 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N , so with small N the answer is fragile
(▶ Appendix)

Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- ▣ Risk-premium PCA and three-pass premia keep the PCA backbone but target pricing (▶ [Appendix](#)); PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#)

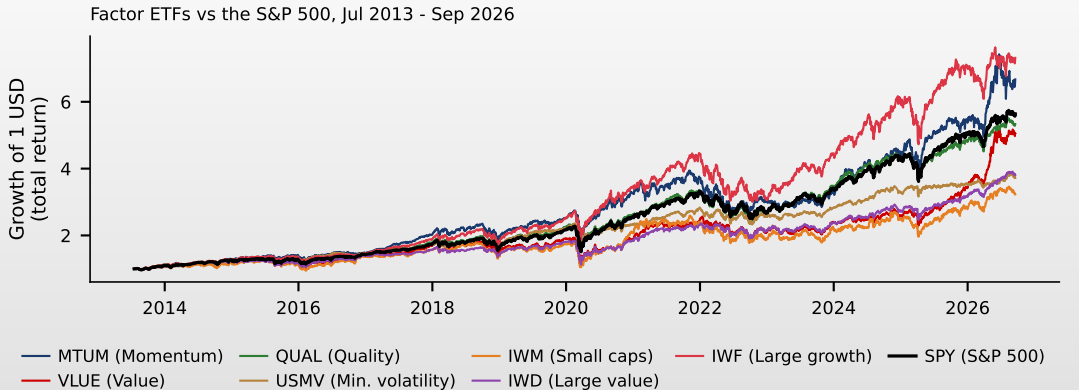
Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of L observed characteristics $\mathbf{z}_{i,t}$ (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- ▣ $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas (“covariances”), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?
- ▣ Result in their US stock data: a few IPCA factors price the cross section better than observable-factor models with as many factors, and $\boldsymbol{\Gamma}_\alpha$ is insignificant: “characteristics are covariances”; crypto-asset factors: [Appendix](#)

Factor ETFs versus the Market (1/2)



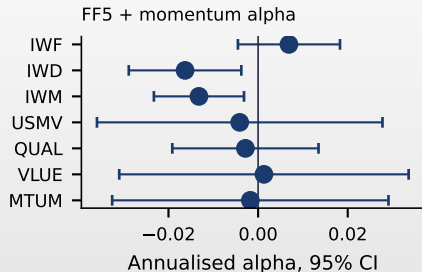
MFM_ch3_factor_etfs

Factor ETFs versus the Market (2/2)

- **Smart beta:** rule-based long-only portfolios that tilt towards factors, sold as ETFs
 - ▶ they keep market exposure and need not replicate an academic long-short factor
- Thirteen years of one realisation: not evidence against the premia on its own
- **CAGR** (Compound Annual Growth Rate) of 1 USD, dividends re-invested, from 18 July 2013
 - ▶ above SPY (14.0%): large growth (IWF) 16.3%, momentum (MTUM) 15.5%
 - ▶ below: quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%
 - ▶ minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%

 MFM_ch3_factor_etfs

Alphas of Factor ETFs



Factor loadings

MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00
	Mkt-RF	SMB	HML	RMW	CMA	MOM

- Daily regressions on FF5 + MOM, Newey–West errors, each ETF on its own trading calendar (2000–2026)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85; IWF -0.28 on HML; USMV market beta 0.74

- Alphas close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%; Russell ETFs IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.55$), consistent with costs and index construction
- R^2 between 0.81 and 0.97: mostly repackaged factor exposure

Timing, Crowding and Costs

- **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
 - ▶ Noisy premia and short histories make timing hard to verify out of sample ([Chapter 8](#))
- **Crowding:** many investors hold the same factor positions
 - ▶ Premia can shrink ([McLean & Pontiff \(2016\)](#)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers ([Daniel & Moskowitz \(2016\)](#))
- **Implementability:** long–short factors ignore trading costs and short-selling fees; ETFs pay them

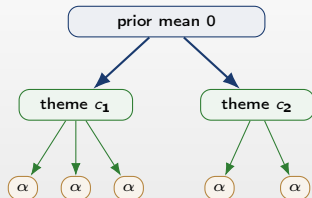
Case Study: Research Questions (1/2)

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
 - ▶ No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
 - ▶ No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Questions**
 - ▶ do published factors have significant CAPM alphas in the original data?
 - ▶ are these alphas significant out of sample and in other countries?
 - ▶ do they remain significant once multiple testing is taken into account?

Case Study: Research Questions (2/2)

- ▣ **Why it matters**
 - ▶ if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- ▣ **A second view of the zoo** (Feng, Giglio & Xiu (2020), [▶ Appendix](#))
 - ▶ most new factors are redundant given the existing ones
 - ▶ the paper instead treats the zoo as correlated evidence on a few economic themes
- ▣ **Link to this chapter**
 - ▶ Jensen's alpha (Section 4) is the object tested
 - ▶ Benjamini–Yekutieli (Section 8) is the frequentist benchmark
 - ▶ Bayesian shrinkage extends the Vasicek beta (Section 3) to alphas

Case Study: Data and Method (1/2)



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries
 - ▶ terciles of each characteristic, capped value weights
 - ▶ long the side with the higher expected return in the original paper
 - ▶ each factor scaled to a 10% annual idiosyncratic volatility
- **Themes**: 13 clusters of US factors
 - ▶ hierarchical clustering on the correlation of CAPM residuals
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
 - ▶ $f_{i,t}$: return of factor i ; r_t^m : market excess return
 - ▶ alphas $\alpha_i = c_j + s_n + w_i$: theme, signal and factor components, prior mean 0

Case Study: Data and Method (2/2)

- One factor: the posterior mean shrinks the estimated alpha (eq. 5)

$$\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}, \quad \kappa = \frac{1}{1 + \sigma^2 / (\tau^2 T)}$$

- ▶ σ : residual volatility; τ : dispersion of the true alphas; T : number of months
- ▶ short, noisy samples are shrunk more, as in Vasicek (Section 3)
- **Replication rate**: share of factors with a significant alpha
 - ▶ Bayesian discovery: posterior mean / posterior s.d. ≥ 1.96
- Full hierarchy and estimation: [▶ Appendix](#)

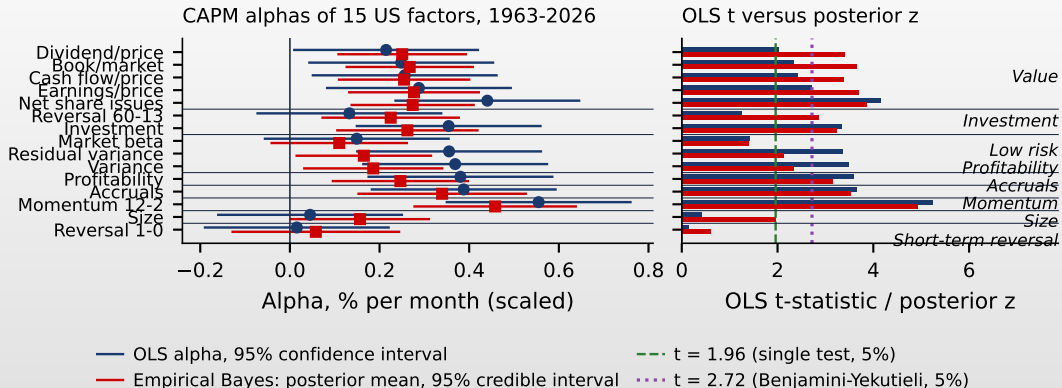
Case Study: Key Results

Replication rates, US (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
This paper, raw returns	55.6%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- Testing alphas instead of raw returns drives most of the gap with Hou, Xue & Zhang (2020)
- Bayesian FDR 0.1%: shrinkage controls false discoveries without the power loss of a higher t hurdle
- **Decay** after the original sample
 - ▶ mean alpha 0.49% a month in sample, 0.26% afterwards (−47%)
 - ▶ slope of post-sample on in-sample alphas 0.26 (Figure 8), against about 0.9 from shrinkage alone
- The strongest factors decay more: arbitrage or data mining (Section 12)
- Complete results: [▶ Appendix](#)

Case Study: Replication on Course Data (1/2)



Case Study: Replication on Course Data (2/2)

- 15 US long–short factors from value-weighted sorts of the [Kenneth French Data Library](#), July 1963 – July 2026 ($T = 757$); CAPM alphas scaled to 10% idiosyncratic volatility; themes of the paper ([authors' cluster labels](#))
- Replication rates: OLS 73.3% (11 of 15), Benjamini–Yekutieli 53.3% (8; critical $t = 2.72$), empirical Bayes 86.7% (13); details:

► [Appendix](#)

- Shrinkage towards the theme, not only towards zero: long-term reversal (investment theme) 0.13% $\rightarrow$ 0.23% a month ($t = 1.25$, posterior $z = 2.85$)
- Failures: market beta and short-term reversal ($z \leq 1.41$); size only just passes ($z = 1.96$), echoing the paper, where size is one of the two themes below 50%

 [MFM_ch3_jkp_replication](#)

Case Study: Limits and Open Questions

▣ Limits of the course version

- ▶ 15 US signals instead of 153 in 93 countries
- ▶ Kenneth French Data Library sort breakpoints instead of non-micro terciles with capped weights
- ▶ one region, so no signal level: $\tau_s = 0$ (τ_s : dispersion of the signal components s_n)
- ▶ The rate depends on the theme structure: with a single common theme the empirical-Bayes rate is 100%

▣ Limits stated by the authors (▶ Appendix)

- ▶ CAPM as the only benchmark; uniform construction of the characteristics; unobserved factors in the prior

▣ Open questions: the paper's data end in 2020

- ▶ did the posterior alphas and the replication rate hold in 2021–2026?
- ▶ can a prior estimated on US factors sharpen short-sample BVB factors?

▣ Data and code of the paper: jpkpfactors.com; course project: Seminar 3, C3

An Open Question and the Discovery Loop

□ Open questions

- ▶ why do factor premia decay after publication: arbitrage, data mining or a change in risk?
- ▶ does the decay differ across factors and markets?

□ Evidence so far

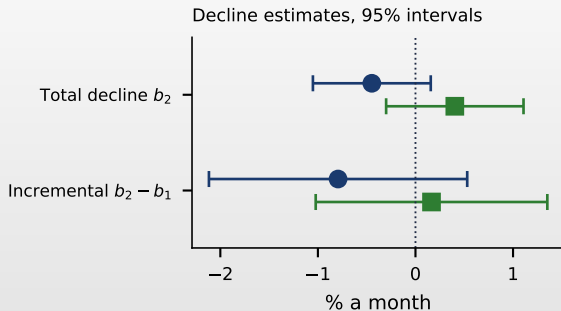
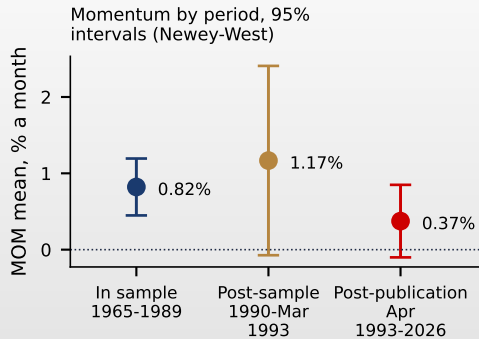
- ▶ [McLean & Pontiff \(2016\)](#): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication; the gap is attributed to publication-informed trading
- ▶ [Jensen, Kelly & Pedersen \(2023\)](#): most factors replicate, but the strongest decay most; [Chen & Zimmermann \(2022\)](#): open data and code for the published predictors

□ One pass of the loop

- ▶ The assistant drafts: a literature table with DOIs, competing hypotheses with their predicted patterns, the regression code, a robustness plan
- ▶ The human fixes before running: the event dates (Jegadeesh–Titman: sample ends 1989, published March 1993), the factor set, the multiple-testing correction, the placebo
- ▶ Checks: every DOI resolves; the chapter's split-sample t -statistics are reproduced first; the data vintage is recorded

□ Reusable prompts and the full checklist: ▶ [Appendix](#)

Mini-Case: Was Momentum Arbitrated Away? (1/2)



— MOM — Mkt-RF (placebo)

Mini-Case: Was Momentum Arbitraged Away? (2/2)

- Design of [McLean & Pontiff \(2016\)](#):
 $MOM_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly
1965–2026, Newey–West errors; post-sample 1990-01 – 1993-03,
post-publication from 1993-04
- PostSample_t , PostPub_t : dummies equal to 1 in those periods; a :
the in-sample mean; b_2 : total decline relative to the in-sample
mean; $b_2 - b_1$: incremental effect of publication
- $a = 0.82\%$ a month ($t = 4.32$); $b_2 = -0.45\%$ ($t = -1.45$,
interval $[-1.05\%, 0.16\%]$): 54% lower; $b_2 - b_1 = -0.79\%$
($t = -1.18$); placebo Mkt-RF $b_2 = +0.40\%$ ($t = 1.12$)
- Close to the 58% of [McLean & Pontiff \(2016\)](#), but one factor
cannot reject $H_0 : b_2 = 0$ at 5%; arbitrage is one interpretation,
not an identified cause

 MFM_ch3_ai_discovery

Your Turn: A Project Seed

▣ Questions

- ▶ do the Fama–French factors and momentum decay after publication?
- ▶ does the decay differ across factors?

▣ Design

- ▶ pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies
- ▶ Mkt-RF as placebo; errors robust to cross-factor correlation and autocorrelation
- ▶ a larger decay where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs)

▣ Replicate first: the split-sample t -statistics and the mini-case estimates

▣ Deliverables

- ▶ a pre-registration note (H_0 , H_1 , dates from each paper, factor set, corrections), committed before the results
- ▶ code as a Quantlet; a report with Newey–West inference, placebo periods and power

▣ Logs: prompts in AI_USE.md; caught errors in AI_ERRORS.md

Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
 2. **Test the pricing restriction jointly**: GRS measures the Sharpe-ratio gain from the test assets; the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, $GRS = 4.20$)
 3. **Estimate premia** in two passes with errors robust to estimated betas and misspecification; a high R^2 on 25 portfolios is a low hurdle
 4. **Guard against the zoo**: multiple-testing hurdles, spanning tests, model comparison by maximum Sharpe ratios
 5. **Replicate**: latent factors and out-of-sample evidence; factor ETFs are cheap factor exposure, not alpha
- Open: why premia decay after publication; how to price with time-varying betas; which factors price assets in small markets such as the BVB

Check Yourself, and Next

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not?
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months?
- Why does a useless factor look priced in pass 2?
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate?

Links to other chapters

- [Chapter 2](#): an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- [Chapter 4](#): factor covariance models and shrinkage for Markowitz
- [Chapters 5–6](#): time-varying volatility, correlation and betas
- [Chapter 8](#): backtesting factor strategies without overfitting
- [Chapter 13](#): machine learning for the cross section of returns

Appendix: GRS under the Null Hypothesis

- Under H_0 and the exact assumptions, $W \sim F(N, T - N - 1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of W : the squared Sharpe-ratio gain expected under H_0 is

$$\mathbb{E}[\widehat{SR}^2(f, R) - \widehat{SR}^2(f)] \approx \frac{N(1 + \widehat{SR}^2(f))}{T - N - 3}$$

- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly), against $0.405^2 - 0.134^2 = 0.146$ observed
- In-sample maximisation over N assets always adds some Sharpe ratio: the test asks whether the gain exceeds this noise level
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$, below the expected $9 \times 1.021/319 = 0.029$

Appendix: Tests with Many Assets

- $\hat{\Sigma} = \hat{E}^\top \hat{E} / T$ has rank at most $T - 2$ (two parameters per regression); $\hat{E}$: the $T \times N$ matrix of residuals
 - ▶ Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$): $\hat{\Sigma}$ is singular, GRS does not exist
 - ▶ Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid $\hat{\Sigma}^{-1}$: standardised sums of squared t -statistics, [Pesaran & Yamagata \(2024\)](#)
- Power enhancement by screening large alphas: [Fan, Liao & Yao \(2015\)](#)

Appendix: Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system: moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top (R_t - X\gamma)$; the sandwich variance with the full Jacobian is the robust one
 - ▶ Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios (RMW: Worked Example slide): s.e. of CMA 2.2% $\rightarrow$ 2.7%, intercept 3.3% $\rightarrow$ 3.8%; HML changes by less than 0.05 percentage points
- Advanced course:  Generalized Method of Moments

Appendix: Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_b e(b)^\top \mathbb{E}[R^e R^{e\top}]^{-1} e(b), \quad e(b) = \mathbb{E}[m_t(b) R_t^e]$$

- ▶ Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$; for gross returns R_t : $e = \mathbb{E}[mR] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
- ▶ Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over two quadrillion factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

Appendix: Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
 - ▶ Step 1: a LASSO of average test-asset returns on their covariances with hundreds of factors selects the controls that price the cross section
 - ▶ Step 2: a second LASSO selects the factors whose covariances explain those of the new factor, so that omitted controls do not bias its SDF coefficient
 - ▶ Step 3: the SDF coefficient of the new factor is estimated with the union of both sets of controls and tested
- Result: most new factors are redundant; only a few add pricing information
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

Appendix: More on the Number of Factors

- ▣ Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- ▣ Kaiser ($\lambda_k > 1$) on a correlation matrix: with N assets and pure noise the eigenvalues spread around 1 with a width that depends on N/T , so the rule has no asymptotic justification
- ▣ Report the range of estimates and the sensitivity to $k_{\max}$, not a single number (Seminar 3, problem B3)

Appendix: Risk-Premium PCA and Three-Pass Premia

- ▣ **RP-PCA** (Risk-Premium PCA, [Lettau & Pelger \(2020\)](#)): principal components of $\frac{1}{T}X^T X + \gamma \bar{x}\bar{x}^T$
 - ▶ X : the $T \times N$ matrix of excess returns, $\bar{x}$: their means, γ : the weight on average returns
 - ▶ $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
 - ▶ In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- ▣ **Three-pass estimator** ([Giglio & Xiu \(2021\)](#)): the premium of an observed factor g_t when the model omits factors
 - ▶ (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- ▣ Both keep PCA's statistical backbone but target pricing, not only variance; IPCA adds characteristics

Appendix: IPCA and Crypto-Asset Factors

- IPCA ([Kelly, Pruitt & Su \(2019\)](#)): characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins ([Liu, Tsyvinski & Wu \(2022\)](#)); dimension reduction on crypto and traditional return features: [Pele et al. \(2023\)](#), [Jheng et al. \(2026\)](#)
- Advanced course:  [Multivariate Statistical Analysis](#)

Appendix: Case Study, Data and Construction (1/2)

- ▣ **Data** (Section II): 153 factors in 93 countries
 - ▶ 138 from the list of [Hou, Xue & Zhang \(2020\)](#), plus 15 from the wider literature
 - ▶ US returns from 1926, most developed markets from 1986
 - ▶ all in US dollars, in excess of the US Treasury bill
- ▣ **Construction**, each country and month
 - ▶ terciles of the characteristic, breakpoints from non-micro stocks
 - ▶ “capped” value weights: market equity winsorised at the NYSE 80th percentile
 - ▶ one-month holding period
 - ▶ at least 5 stocks per leg and 60 monthly observations
- ▣ **Themes**: 13 clusters of US factors (Section II.B)
 - ▶ hierarchical clustering, distance 1— correlation of CAPM residuals, Ward linkage

Appendix: Case Study, Data and Construction (2/2)

- **Model (Section I):** $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
 - ▶ alphas $\alpha_i = c_j + s_n + w_i$ (theme j , signal n , factor i ; eq. 21), prior mean 0
 - ▶ Prior variance $\Omega = MM^\top \tau_c^2 + ZZ^\top \tau_s^2 + I \tau_w^2$ (eq. 23); posterior mean $(\Omega^{-1} + T\Sigma^{-1})^{-1} T\Sigma^{-1} \hat{\alpha}$ (Proposition 4)
 - ▶ M, Z: 0/1 matrices linking each factor to its theme and signal; $\tau_c^2, \tau_s^2, \tau_w^2$: variances of the three alpha components; Σ : residual covariance
 - ▶ Empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\alpha} \sim N(0, \Omega + \hat{\Sigma}/T)$ (Section I.C.2)
- **Replication rate**
 - ▶ share of factors significant in the original paper whose alpha is significant here

Appendix: Case Study, Complete Results

- Replication rate of the factors that were significant in the original papers: 61.3% (Figure 1)
- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size: not an artefact of micro-caps
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

Appendix: Case Study, the Course Replication in Detail

- ▣ Benjamini–Yekutieli is the strictest, as in the US rates of the paper (Figure 1)
- ▣ Net share issues move towards their theme: 0.44% $\rightarrow$ 0.27% a month
- ▣ Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- ▣ **Prior dispersion:** $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month
 - ▶ against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- ▣ **Themes:** the 15 signals fall into 8 of the paper's 13 themes
 - ▶ 5 in value, 3 in low risk, 2 in investment

Appendix: Case Study, Limits Stated by the Authors and Extensions

▣ Limits stated by the authors

- ▶ CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
- ▶ Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
- ▶ Unobserved factors bias the prior: with the scenarios of [Harvey, Liu & Zhu \(2016\)](#) the US rate falls to 81.5% and 79.8% (Figure 10)

▣ Complete the course replication (Seminar 3, C3)

- ▶ justify each sign from the original paper
- ▶ compare the dendrogram with the paper's themes
- ▶ add the Bayesian FDR and the post-sample slope (Figure 8)

▣ Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; check each headline number against the abstract

□ Hypothesis

- ▶ Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.

□ Data and code

- ▶ Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.
- ▶ Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).

▣ Critique

- ▶ Referee this: one factor, one publication date, $t = -1.45$. What is the power to detect a 58% decay, and why does pooling many anomalies change it?

▣ Report

- ▶ In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
- ▶ Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant enumerates specifications; the human fixes which ones count before running them

Appendix: What the Researcher Must Check

- **Wrong event date:** sample end, working paper and journal issue are different dates
 - ▶ test: read them from the paper itself; Jegadeesh–Titman: sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay:** fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns:** Newey–West errors; rerun without 2009
- **Data vintages:** the factor library is revised retroactively
 - ▶ test: record the download date; reproduce the chapter's split-sample t -statistics
- **Misquoted literature:** a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power:** a single factor rarely rejects; report the confidence interval of the decay, not only its sign

Appendix: Further Study and Code

Advanced courses on quantinar.com

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- ▣ Factor data: [Kenneth French Data Library](#)

References (1/6)

- ▣ Ahn, S.C., Horenstein, A.R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81(3), 1203–1227.
- ▣ Anatolyev, S., Mikusheva, A. (2022). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, 229(1), 103–126.
- ▣ Ang, A., Liu, J., Schwarz, K. (2020). Using stocks or portfolios in tests of factor models. *Journal of Financial and Quantitative Analysis*, 55(3), 709–750.
- ▣ Bai, J., Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- ▣ Barillas, F., Shanken, J. (2018). Comparing asset pricing models. *Journal of Finance*, 73(2), 715–754.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- ▣ Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- ▣ Blume, M.E. (1971). On the assessment of risk. *Journal of Finance*, 26(1), 1–10.
- ▣ Bryzgalova, S., Huang, J., Julliard, C. (2023). Bayesian solutions for the factor zoo: we just ran two quadrillion models. *Journal of Finance*, 78(1), 487–557.

References (2/6)

- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Carhart, M.M. (1997). *On persistence in mutual fund performance*. *Journal of Finance*, 52(1), 57–82.
- ▣ Chen, A.Y., Zimmermann, T. (2022). *Open source cross-sectional asset pricing*. *Critical Finance Review*, 11(2), 207–264.
- ▣ Cochrane, J.H. (2011). *Presidential address: discount rates*. *Journal of Finance*, 66(4), 1047–1108.
- ▣ Daniel, K., Moskowitz, T.J. (2016). *Momentum crashes*. *Journal of Financial Economics*, 122(2), 221–247.
- ▣ Fama, E.F., French, K.R. (1992). *The cross-section of expected stock returns*. *Journal of Finance*, 47(2), 427–465.
- ▣ Fama, E.F., French, K.R. (1993). *Common risk factors in the returns on stocks and bonds*. *Journal of Financial Economics*, 33(1), 3–56.
- ▣ Fama, E.F., French, K.R. (2015). *A five-factor asset pricing model*. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Fama, E.F., MacBeth, J.D. (1973). *Risk, return, and equilibrium: empirical tests*. *Journal of Political Economy*, 81(3), 607–636.
- ▣ Fan, J., Liao, Y., Yao, J. (2015). *Power enhancement in high-dimensional cross-sectional tests*. *Econometrica*, 83(4), 1497–1541.

References (3/6)

- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- ▣ Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- ▣ Giglio, S., Xiu, D. (2021). Asset pricing with omitted factors. *Journal of Political Economy*, 129(7), 1947–1990.
- ▣ Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4), 1029–1054.
- ▣ Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- ▣ Härdle, W.K., Simar, L. (2019). *Applied Multivariate Statistical Analysis*, 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- ▣ Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

References (4/6)

- ▣ Hou, K., Xue, C., Zhang, L. (2015). [Digesting anomalies: an investment approach](#). *Review of Financial Studies*, 28(3), 650–705.
- ▣ Hou, K., Xue, C., Zhang, L. (2020). [Replicating anomalies](#). *Review of Financial Studies*, 33(5), 2019–2133.
- ▣ Jagannathan, R., Wang, Z. (1998). [An asymptotic theory for estimating beta-pricing models using cross-sectional regression](#). *Journal of Finance*, 53(4), 1285–1309.
- ▣ Jegadeesh, N., Titman, S. (1993). [Returns to buying winners and selling losers: implications for stock market efficiency](#). *Journal of Finance*, 48(1), 65–91.
- ▣ Jensen, M.C. (1968). [The performance of mutual funds in the period 1945–1964](#). *Journal of Finance*, 23(2), 389–416.
- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). [Is there a replication crisis in finance?](#). *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jheng, S.-L., Găman, Ș., Conda, A., Pele, D.T., Härdle, W.K., Teng, H.-W., Osterrieder, J. (2026). [Crypto currency returns](#). SSRN Working Paper.
- ▣ Kan, R., Robotti, C., Shanken, J. (2013). [Pricing model performance and the two-pass cross-sectional regression methodology](#). *Journal of Finance*, 68(6), 2617–2649.
- ▣ Kan, R., Zhang, C. (1999). [Two-pass tests of asset pricing models with useless factors](#). *Journal of Finance*, 54(1), 203–235.
- ▣ Kelly, B.T., Pruitt, S., Su, Y. (2019). [Characteristics are covariances: a unified model of risk and return](#). *Journal of Financial Economics*, 134(3), 501–524.

References (5/6)

- ▣ Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- ▣ Kozak, S., Nagel, S., Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.
- ▣ Lettau, M., Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- ▣ Lewellen, J., Nagel, S., Shanken, J. (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics*, 96(2), 175–194.
- ▣ Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- ▣ Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability?. *Journal of Finance*, 71(1), 5–32.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92(4), 1004–1016.

References (6/6)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Pesaran, M.H., Yamagata, T. (2024). Testing for alpha in linear factor pricing models with a large number of securities. *Journal of Financial Econometrics*, 22(2), 407–460.
- ▣ Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Roll, R. (1977). A critique of the asset pricing theory's tests. Part I. *Journal of Financial Economics*, 4(2), 129–176.
- ▣ Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- ▣ Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- ▣ Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.
- ▣ Vasicek, O.A. (1973). A note on using cross-sectional information in Bayesian estimation of security betas. *Journal of Finance*, 28(5), 1233–1239.
- ▣ Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.